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(54) **REDUCING INCONVENIENCE TO SURROUNDING ROAD USERS CAUSED BY STOPPED AUTONOMOUS VEHICLES**

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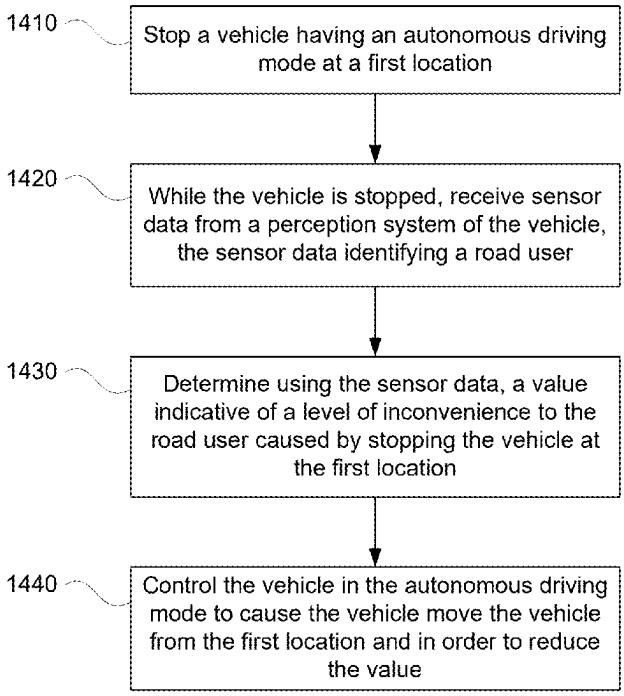
(57) **ABSTRACT**

Aspects of the disclosure provide for reducing inconvenience to other road users caused by stopped autonomous vehicles. As an example, a vehicle having an autonomous driving mode may be stopped at a first location. While the vehicle is stopped, sensor data is received from a perception system of the vehicle. The sensor data may identify a road user. Using the sensor data, a value indicative of a level of inconvenience to the road user caused by stopping the vehicle at the first location may be determined. The vehicle is controlled in the autonomous driving mode to cause the vehicle to move from the first location and in order to reduce the value.



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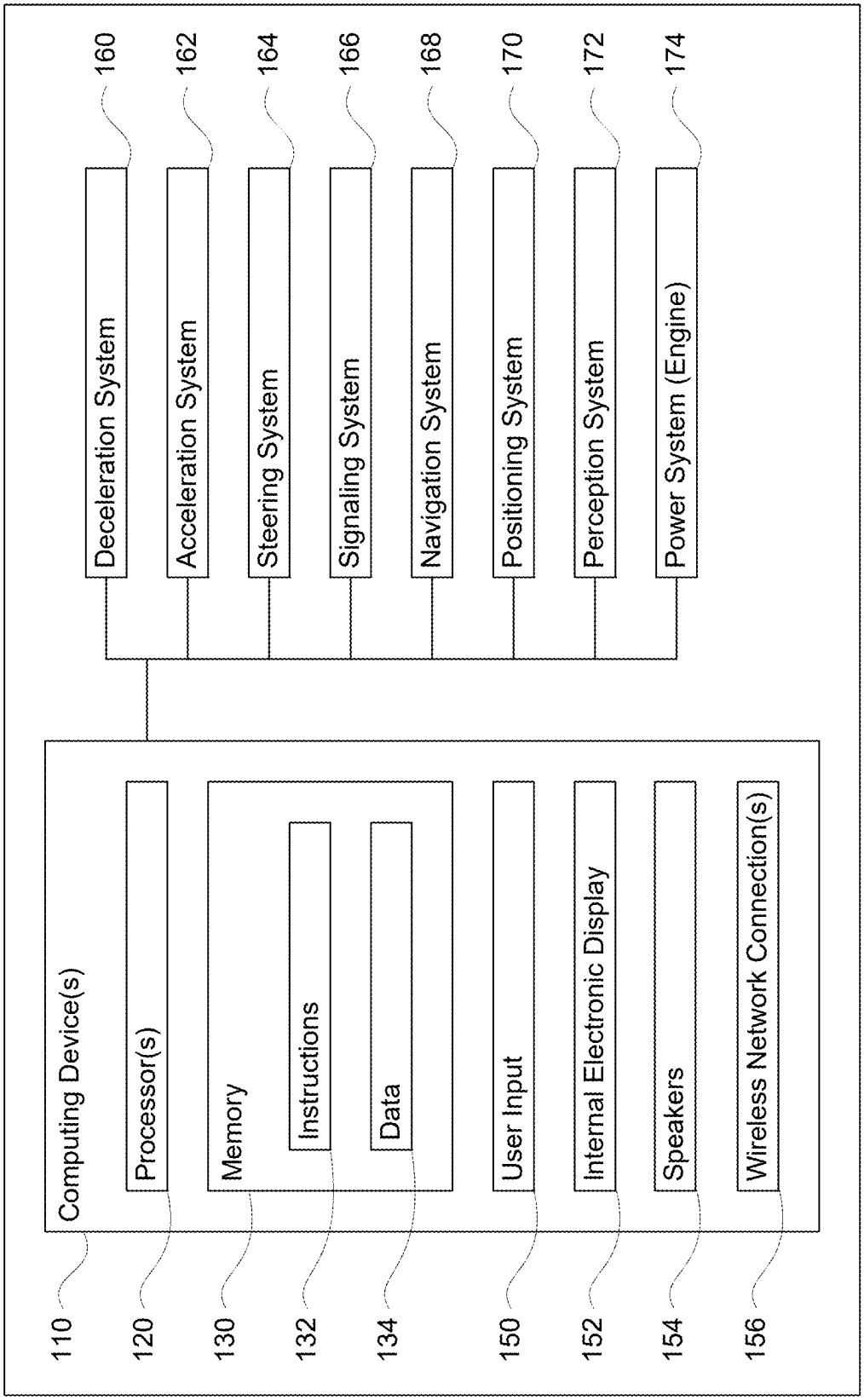
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110
Processor(s)
120
Memory
130
Instructions
132
Data
134
User Input
150
Internal Electronic Display
152
Speakers
154
Wireless Network Connection(s)
156
Deceleration System
160
Acceleration System
162
Steering System
164
Signaling System
166
Navigation System
168
Positioning System
170
Perception System
172
Power System (Engine)
174

100

FIGURE 1

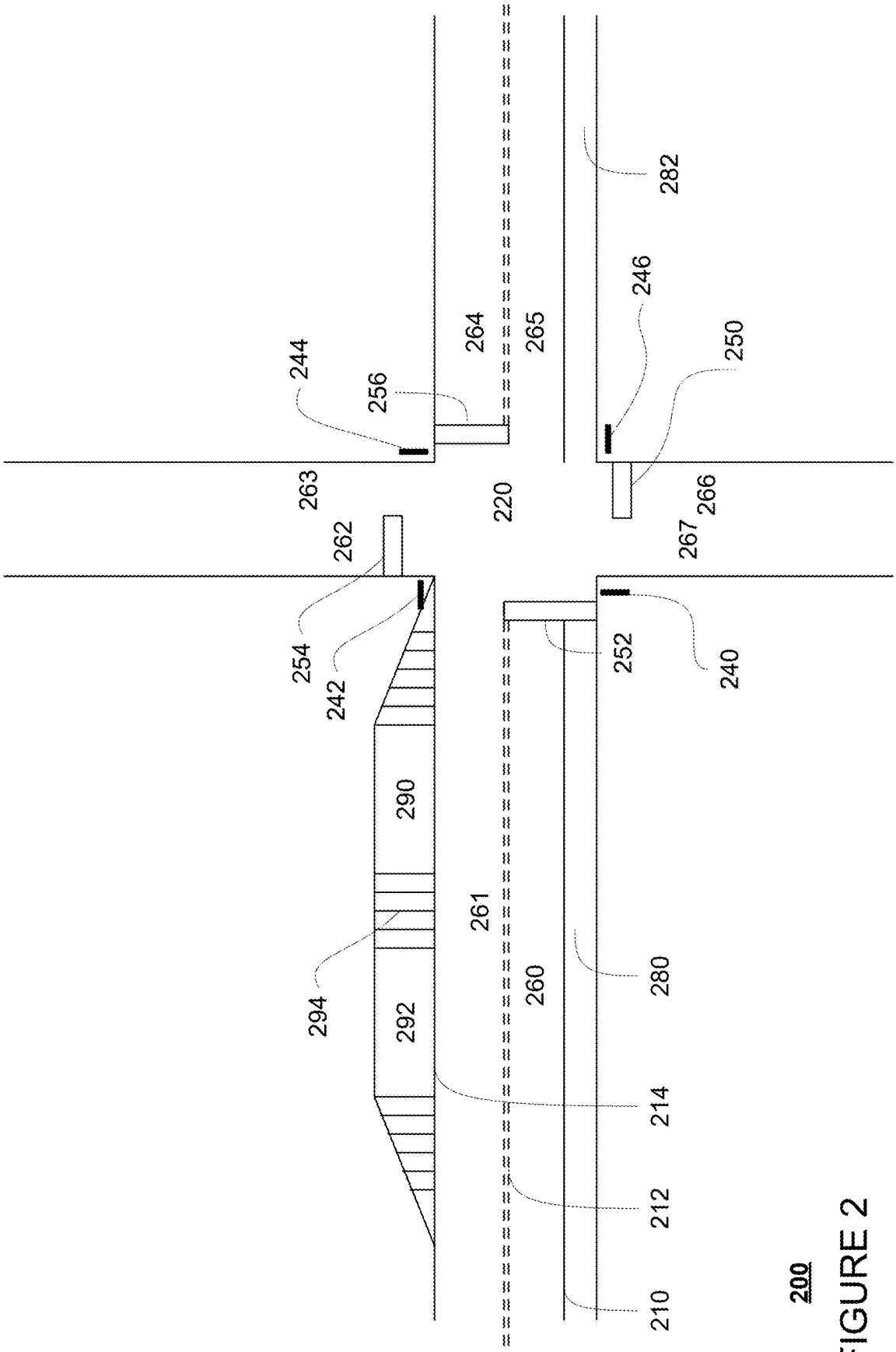
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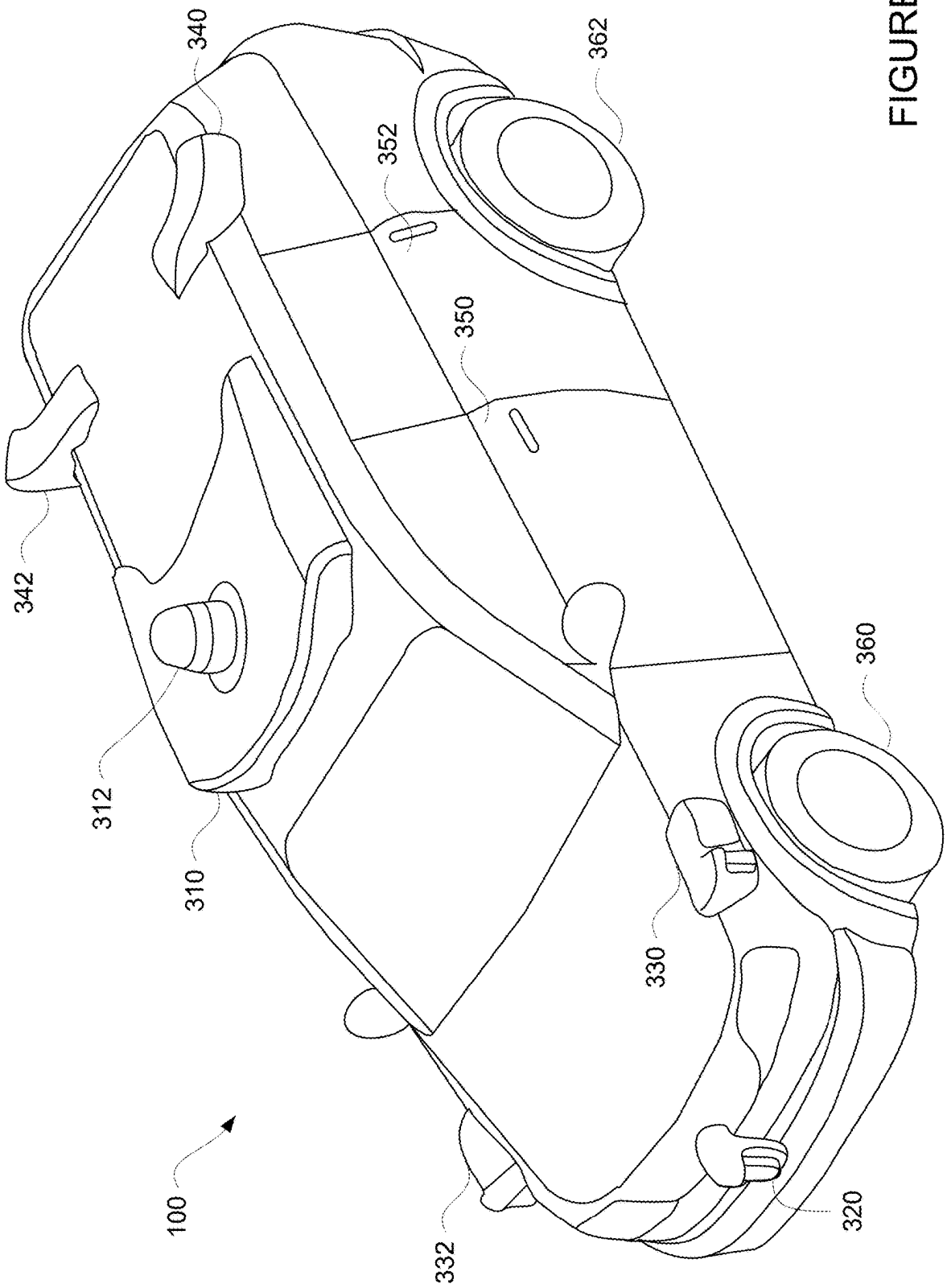
FIGURE 2



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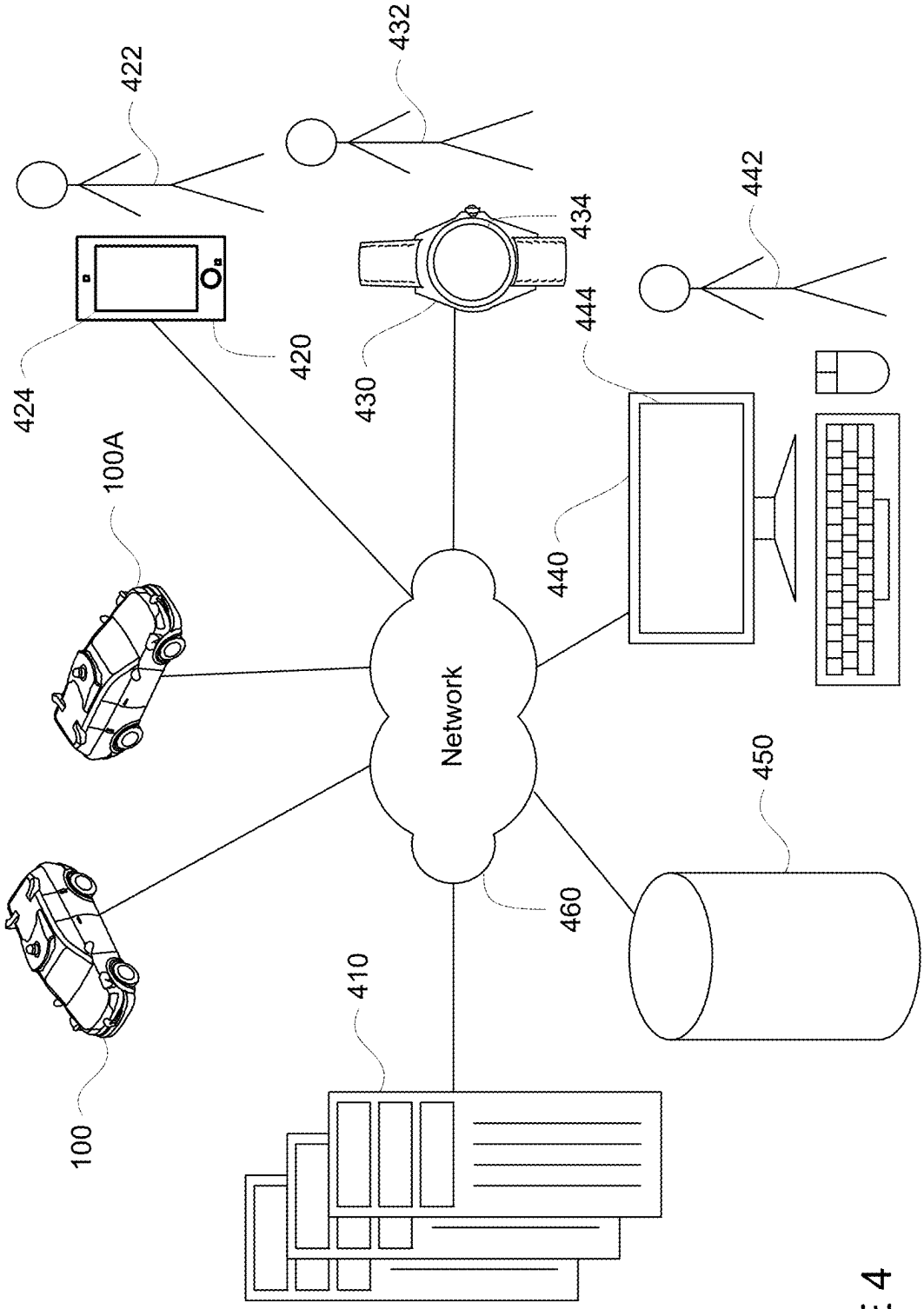
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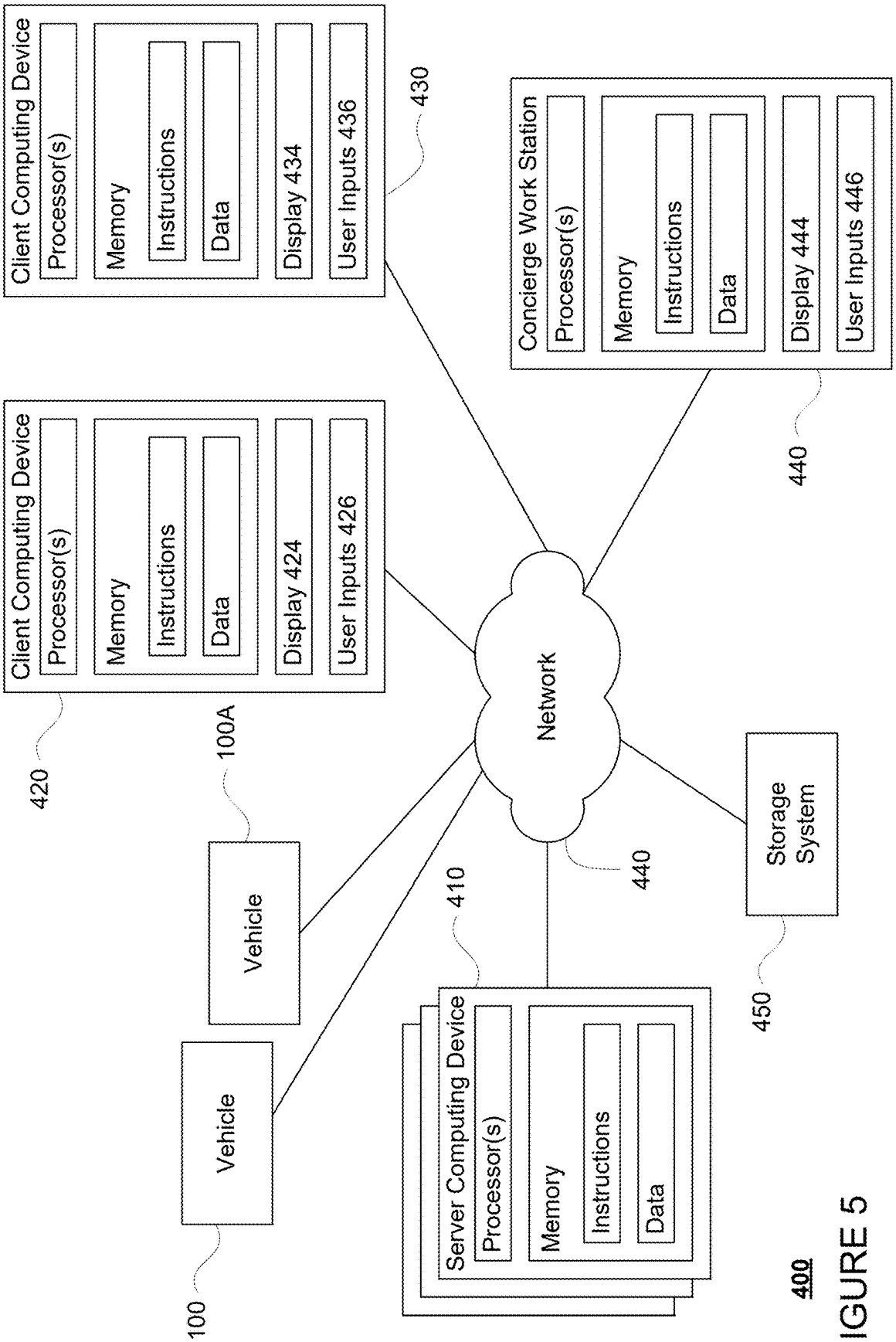
FIGURE 4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Processor(s)
Memory
Instructions
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Display 434
User Inputs 436
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Processor(s)
Memory
Instructions
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Display 424
User Inputs 426
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Concierge Work Station
Processor(s)
Memory
Instructions
Data
Display 444
User Inputs 446
440
Network
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Storage System
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Vehicle
100A
Vehicle
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Server Computing Device
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Memory
Instructions
Data
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400

FIGURE 5

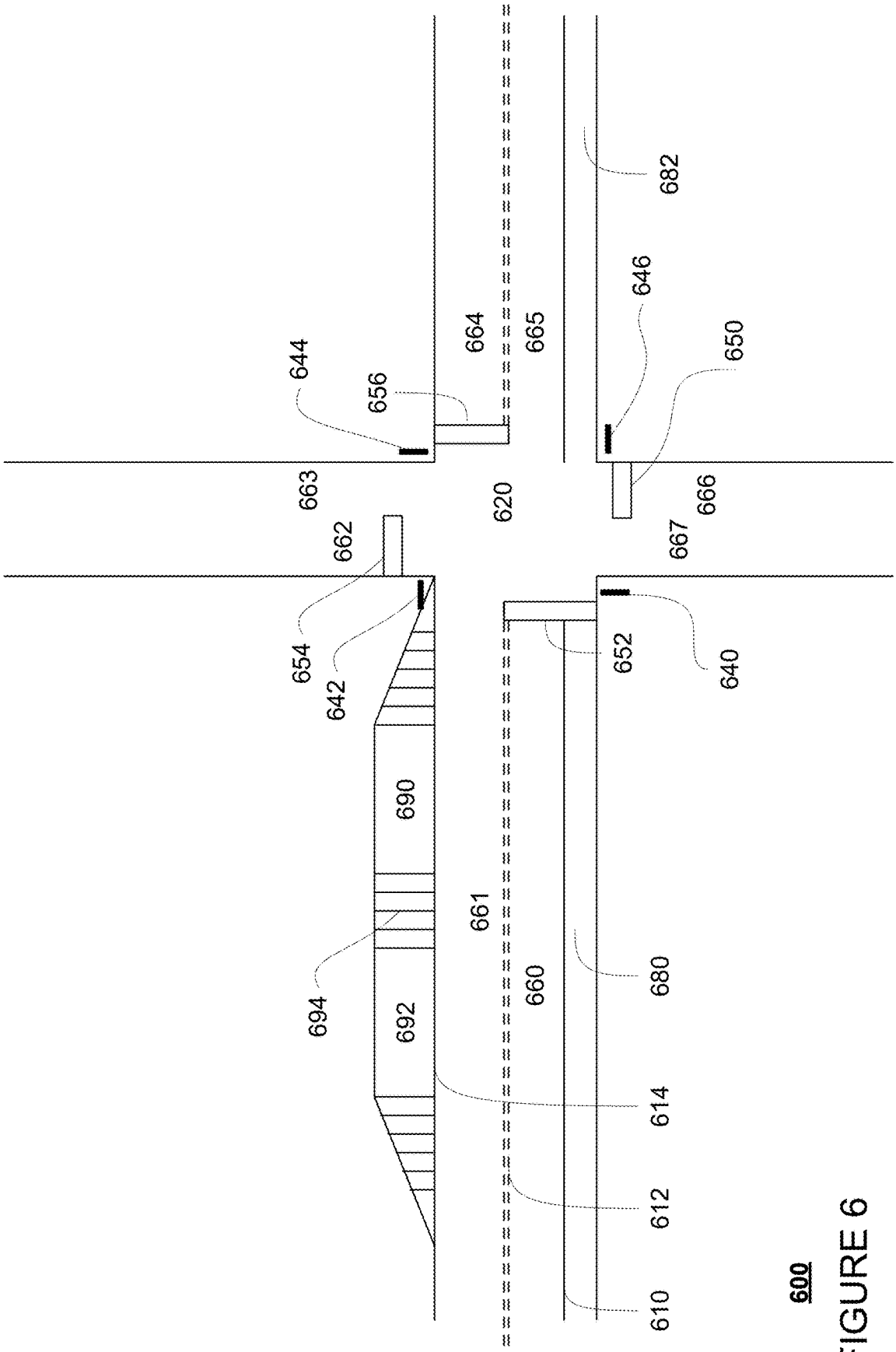
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FIGURE 6

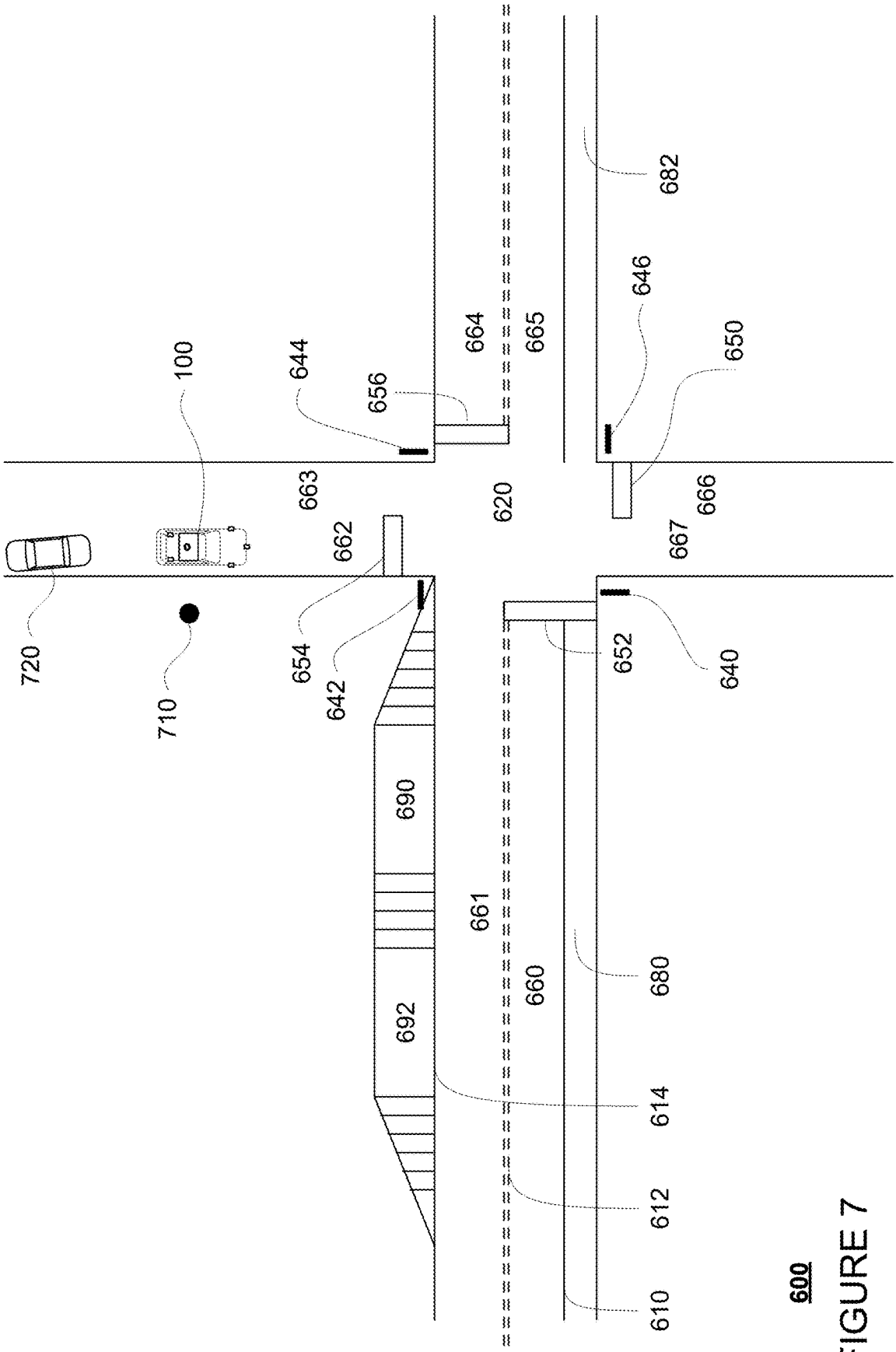
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FIGURE 7

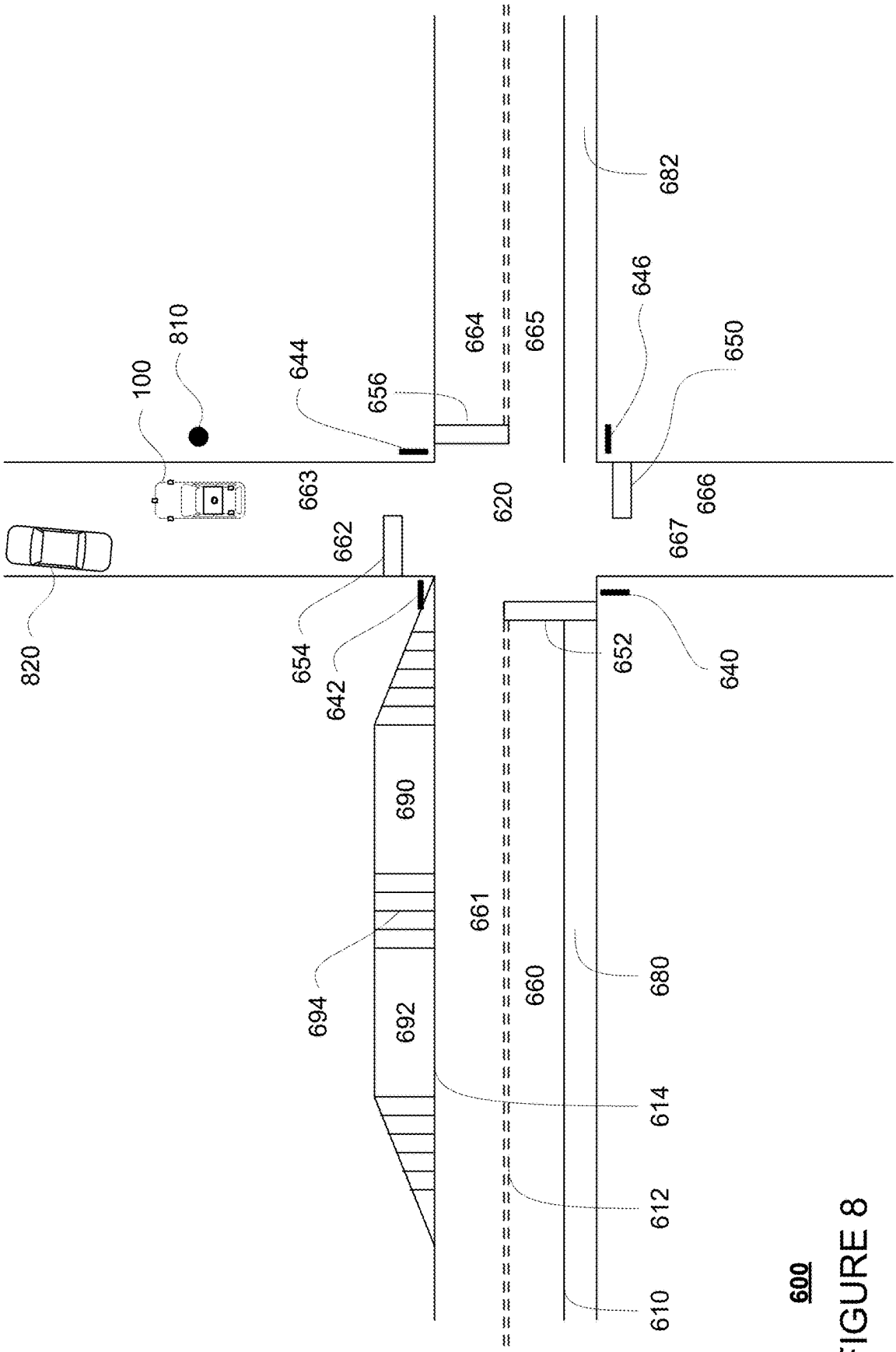
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FIGURE 8

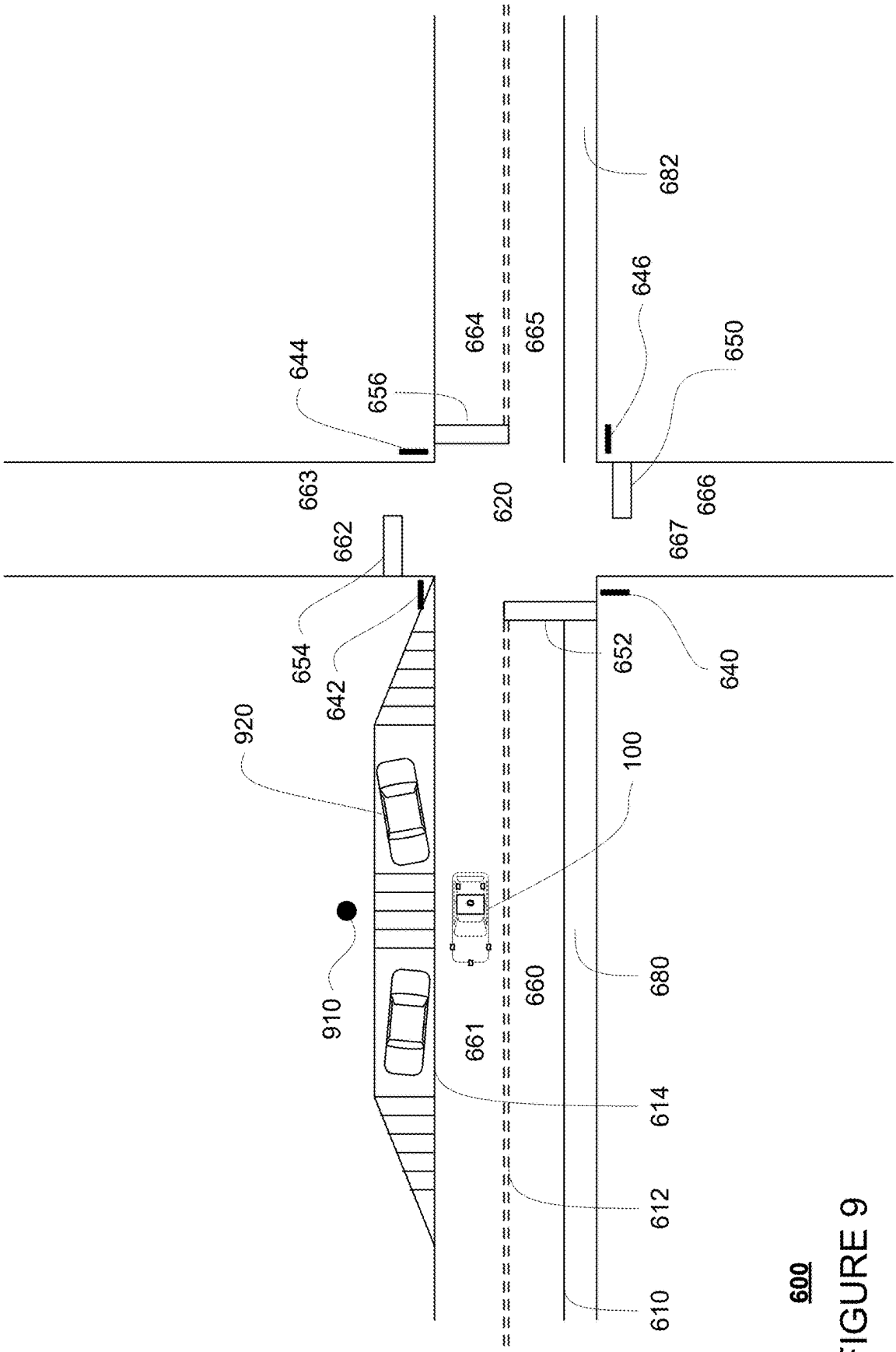
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FIGURE 9

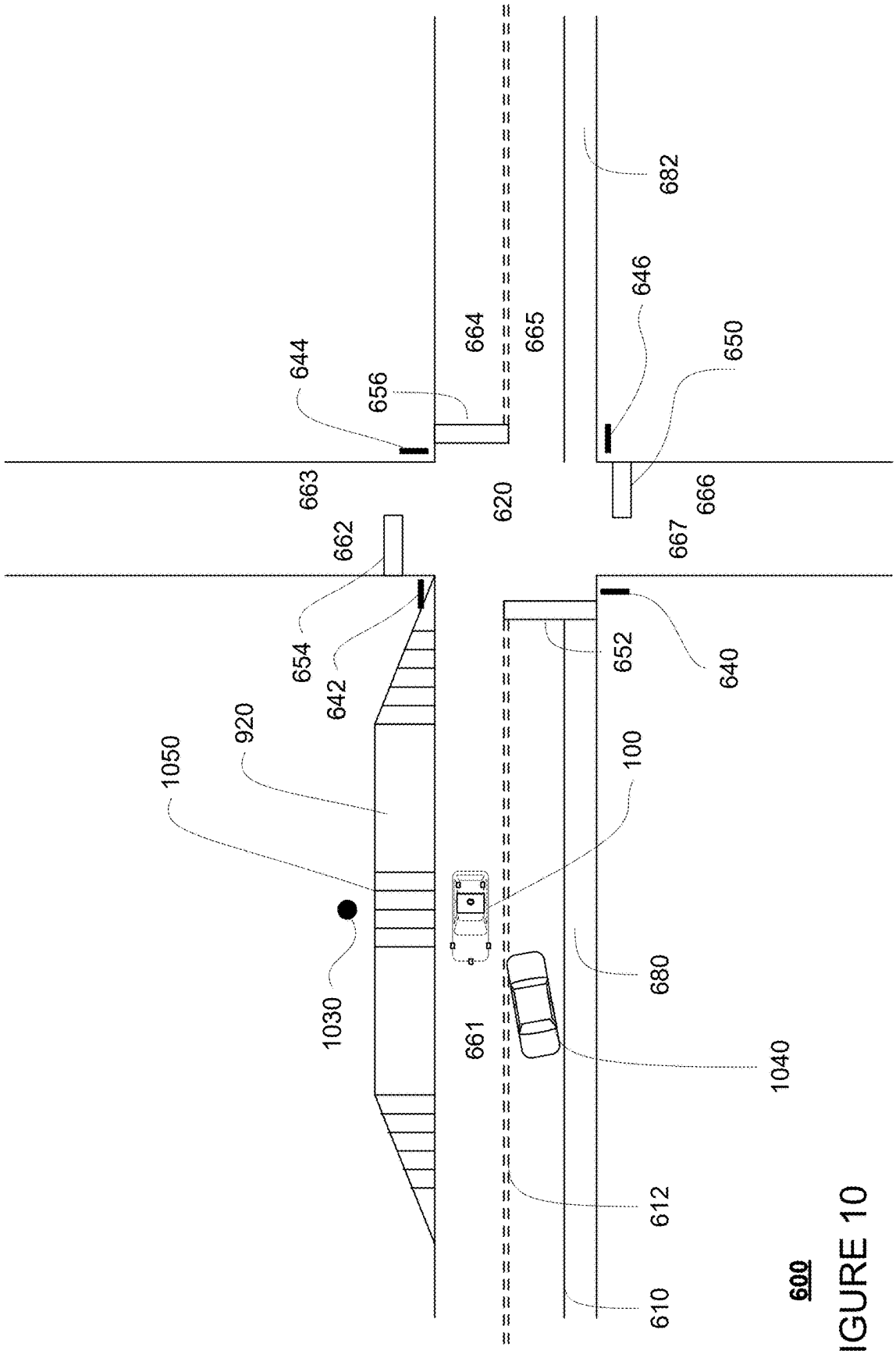
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FIGURE 10

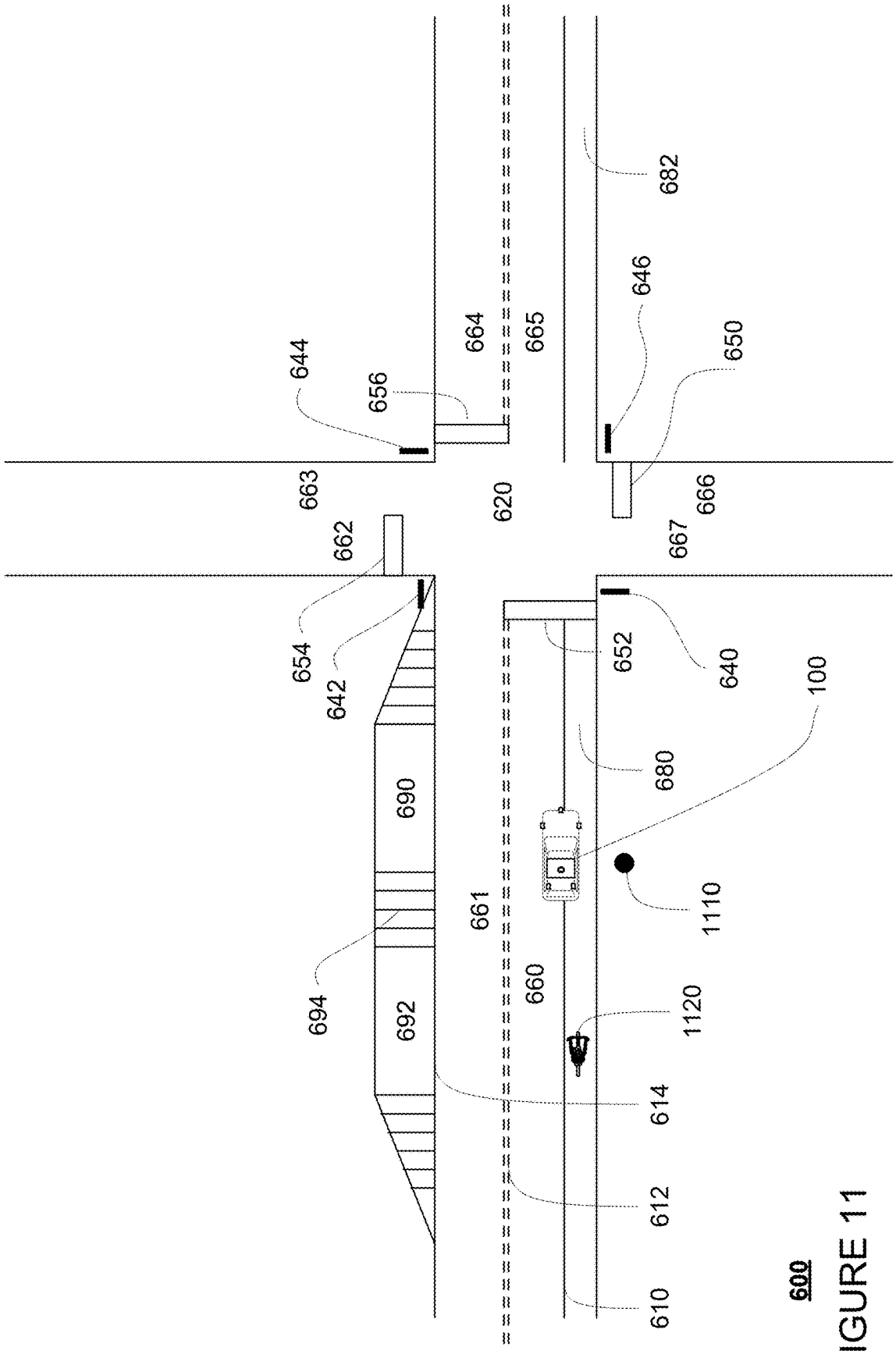
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FIGURE 11

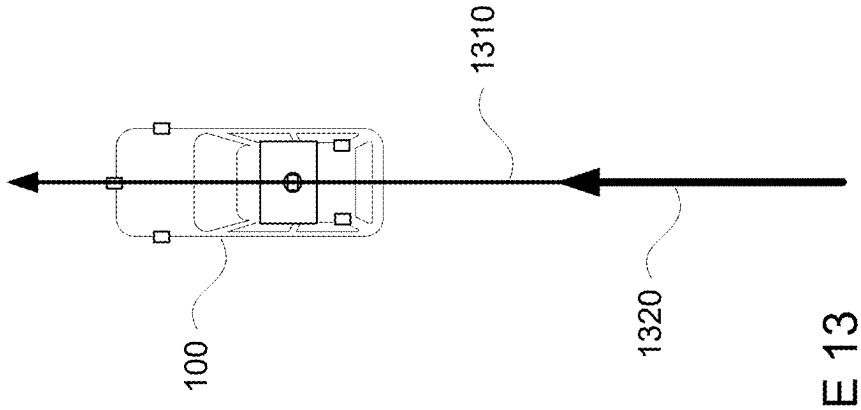

FIGURE 13

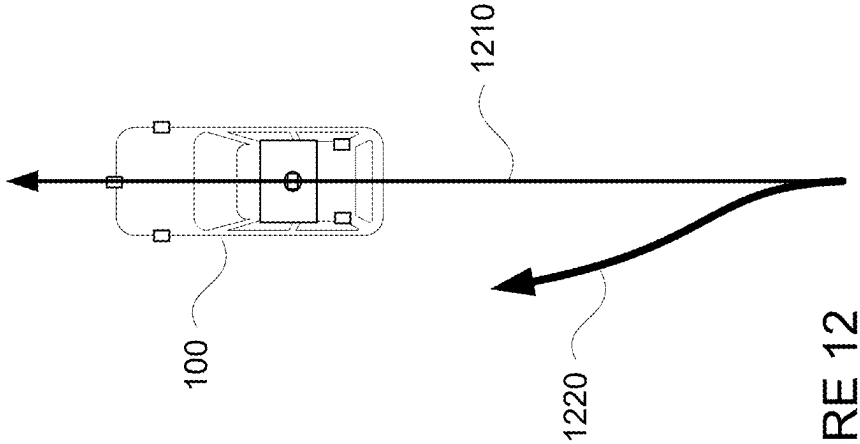

FIGURE 12

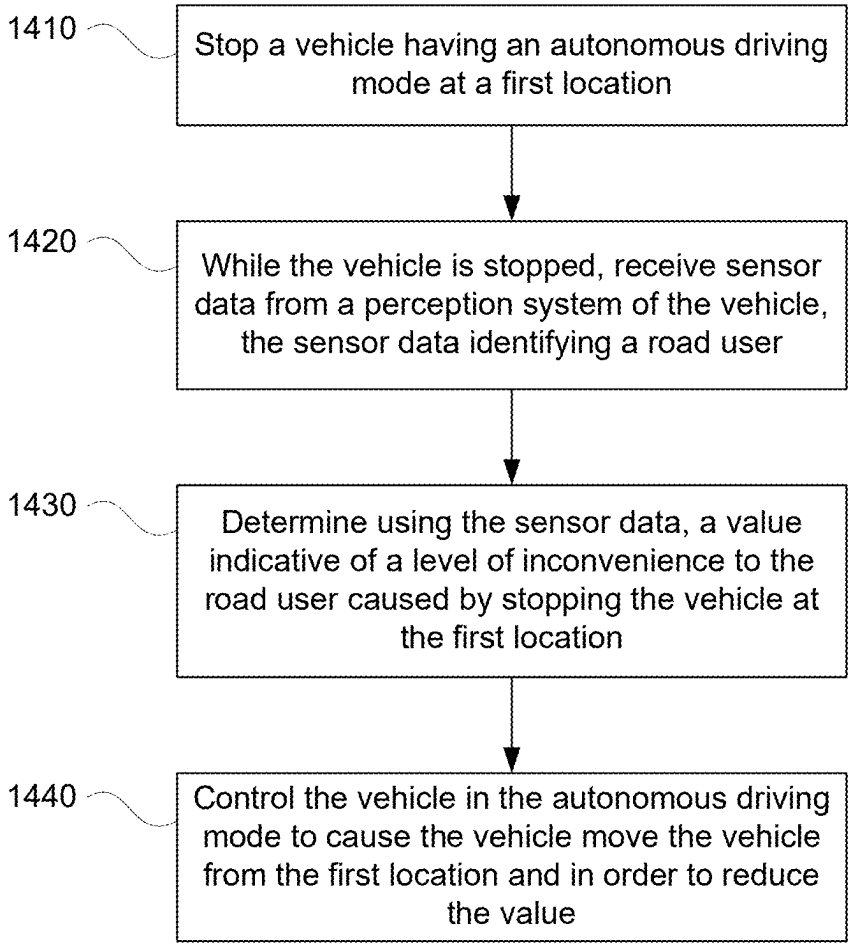

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FIGURE 14

REDUCING INCONVENIENCE TO SURROUNDING ROAD USERS CAUSED BY STOPPED AUTONOMOUS VEHICLES

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation of U.S. patent application Ser. No. 18/074,901, filed Dec. 5, 2022, which is a continuation of U.S. patent application Ser. No. 17/177,278, filed Feb. 17, 2021, issued as U.S. Pat. No. 11,543,823, which is a continuation of U.S. patent application Ser. No. 16/141,467, filed Sep. 25, 2018, issued as U.S. Pat. No. 10,955,848, the entire disclosures of which are incorporated herein by reference.

BACKGROUND

[0002] Autonomous vehicles, such as vehicles that do not require a human driver, can be used to aid in the transport of passengers or items from one location to another. Such vehicles may operate in a fully autonomous mode where passengers may provide some initial input, such as a pickup or destination location, and the vehicle maneuvers itself to that location.

[0003] When a person (or user) wants to be physically transported between two locations via a vehicle, they may use any number of transportation services. To date, these services typically involve a human driver who is given dispatch instructions to a location to pick up the user. In many cases, the human driver and the user are able to arrange an exact location for the user to be picked up. In addition, when waiting for passengers are stopped and waiting for a passenger, human drivers are typically able to recognize when they are inconveniencing other road users such as other vehicles, pedestrians, bicyclists, etc. in their environment. For instance, a vehicle can block or inconvenience other vehicles behind it, vehicles approaching from an adjacent lane, vehicles attempting to park or unpark, bicyclists and/or bicycle lanes, pedestrians attempting to cross, and so on. As such, a human driver may be able to determine how to best reduce inconvenience to other drivers. This is not readily achievable in the case of autonomous vehicles which do not have a human driver.

BRIEF SUMMARY

[0004] One aspect of the disclosure provides a method of reducing inconvenience to other road users for autonomous vehicles. The method includes stopping, by one or more processors, a vehicle having an autonomous driving mode at a first location; while the vehicle is stopped, receiving, by one or more processors, sensor data from a perception system of the vehicle, the sensor data identifying a road user; determining, by the one or more processors, using the sensor data, a value indicative of a level of inconvenience to the road user caused by stopping the vehicle at the first location; and controlling, by the one or more processors, the vehicle in the autonomous driving mode to cause the vehicle to move from the first location and in order to reduce the value.

[0005] In one example, the sensor data indicates observed behavior for the other road user, and the method further includes receiving output of a behavior model predicting future behavior of the other road user and comparing the future behavior to the observed behavior, and wherein the determining the value is further based on the comparison. In another example, the sensor data indicates observed behavior for the other road user, and the method further includes receiving first output of a behavior model predicting a first future behavior of the other road user, after receiving the first output, receiving second output of the behavior model predicting a second future behavior of the road user; and comparing the first future behavior to the second future behavior, and wherein the determining the value is further based on the comparison. In this example, the second output further indicates that the second future behavior is responsive to the vehicle, and determining the value is further based on the indication that the second future behavior is responsive to the vehicle. In another example, determining the value includes inputting the sensor data into a model which outputs the value. In this example, the method also includes determining that the vehicle is inconveniencing the road user based on output from a behavior model predicting a first future behavior of the other road user, and wherein the sensor data is input into the model based on the determination that the vehicle is inconveniencing the road user. In another example, controlling the vehicle is based on whether the road user is in a lane adjacent to a lane of the vehicle. In another example, controlling the vehicle is based on whether the road user is in a lane of the vehicle and behind the vehicle. In another example, the vehicle is stopped in the first location in order to wait for a passenger, and the method also includes sending a message to a client computing device of the passenger based on the controlling. In another example, the method also includes determining that the value meets a threshold, and wherein controlling the vehicle is further based on the determination that the value meets the threshold.

[0006] Another aspect of the disclosure provides a system for reducing inconvenience to other road users caused by stopped autonomous vehicles. The system includes one or more processors configured to stop a vehicle having an autonomous driving mode at a first location; while the vehicle is stopped, receive sensor data from a perception system of the vehicle, the sensor data identifying a road user; determine using the sensor data, a value indicative of a level of inconvenience to the road user caused by stopping the vehicle at the first location; and controlling, by the one or more processors, the vehicle in the autonomous driving mode to cause the vehicle to move from the first location and in order to reduce the value.

[0007] In one example, the sensor data indicates observed behavior for the other road user, and the one or more processors are also configured to receive output of a behavior model predicting future behavior of the other road user and compare the future behavior to the observed behavior, and wherein determining the value is further based on the comparison. In another example, the sensor data indicates observed behavior for the other road user, and the one or more processors are also configured to receive first output of a behavior model predicting a first future behavior of the other road user; after receiving the first output, receive second output of the behavior model predicting a second future behavior of the road user; and compare the first future behavior to the second future behavior, and wherein determining the value is also based on the comparison. In this example, second output also indicates that the second future behavior is responsive to the vehicle, and the one or more processors are also configured to determine the value also based on the indication that the second future behavior is

responsive to the vehicle. In another example, the one or more processors are also configured to determine the value includes inputting the sensor data into a model which outputs the value. In this example, the one or more processors are also configured to determine that the vehicle is inconveniencing the road user based on output from a behavior model predicting a first future behavior of the other road user, and the sensor data is input into the model based on the determination that the vehicle is inconveniencing the road user. In another example, the one or more processors are also configured to control the vehicle also based on whether the road user is in a lane adjacent to a lane of the vehicle. In another example, the one or more processors are also configured to control the vehicle also based on whether the road user is in a lane of the vehicle and behind the vehicle. In another example, the vehicle is stopped in the first location in order to wait for a passenger, and the one or more processors are further configured to send a message to a client computing device of the passenger based on the controlling. In another example, the one or more processors are further configured to determine that the value meets a threshold, and to control the vehicle further based on the determination that the value meets the threshold.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a functional diagram of an example vehicle in accordance with aspects of the disclosure.

[0009] FIGS. **2** is an example representation of map information in accordance with aspects of the disclosure.

[0010] FIG. **3** is an example external view of a vehicle in accordance with aspects of the disclosure.

[0011] FIG. **4** is an example pictorial diagram of a system in accordance with aspects of the disclosure.

[0012] FIG. **5** is an example functional diagram of a system in accordance with aspects of the disclosure.

[0013] FIG. **6** is a view of a section of roadway in accordance with aspects of the disclosure.

[0014] FIGS. **7-11** are examples of various situations in accordance with aspects

[0015] of the disclosure.

[0016] FIG. **12** is an example comparison of an actual trajectory and a predicted trajectory for a road user in accordance with aspects of the disclosure.

[0017] FIG. **13** is another example comparison of an actual trajectory and a predicted trajectory for a road user in accordance with aspects of the disclosure.

[0018] FIG. **14** is flow diagram in accordance with aspects of the disclosure.

DETAILED DESCRIPTION

Overview

[0019] The technology relates to enabling autonomous vehicles to make independent determinations as to whether to move from a pick up location when waiting for a passenger to arrive. This in turn, may reduce inconvenience to other road users in the surrounding environment of an autonomous vehicle stopped and waiting for one or more passengers. For instance, as noted above, a vehicle can block or inconvenience other vehicles behind it, vehicles approaching from an adjacent lane, vehicles attempting to park or unpark, bicyclists and/or bicycle lanes, pedestrians attempting to cross, and so on.

[0020] As one approach, when stopped, an autonomous vehicle may request assistance at increasing time intervals from remote operators as to whether the autonomous should move along based on whether the vehicle is blocking or otherwise inconveniencing other road users. In response, the remote operator may review the vehicle's surroundings and instruct the autonomous vehicle to stay put or move along depending on the flow of traffic around the vehicle. Of course, this back-and-forth can take time, which can cause annoyance and inconvenience to other road users. To reduce or even eliminate the need for the autonomous vehicles to call a remote operator in such situations, the vehicle's computing devices may identify situations in which the vehicle is inconveniencing other drivers and thereafter attempt to reduce that inconvenience.

[0021] In order to do so, the vehicle's computing devices may receive sensor data from the vehicle's perception system. The perception system may include various sensors and may be configured to detect and identify objects, such as other road users, in the vehicle's environment. This information may then be used to determine whether the vehicle is inconveniencing other road users, such as by blocking traffic or causing other road users to have to move around the vehicle. For instance, the vehicle's computing devices may determine whether the vehicle is preventing other vehicles, pedestrians, and bicyclists from proceeding or causing those road users to have to drive around the vehicle.

[0022] The vehicle's computing devices may receive output from various behavior prediction models. For instance, for each road user of a given type detected by the vehicle's perception system, the vehicle's computing devices may use a corresponding behavior prediction model for that type of road user to estimate a future behavior of that road user. In some instances, these behavior models may also identify whether the other road user's behavior is responsive to the vehicle. If the predictions change over time or if a prediction does not sufficiently correspond with the actual behavior of the other road user as the other road user approaches the vehicle and these changes or differences are caused by the vehicle, this may indicate that the vehicle is inconveniencing the other road user.

[0023] In order to allow the vehicle's computing devices to quickly assess the inconvenience, a model may be used to provide an inconvenience value for only those other road users detected by the vehicle's perception system as being inconvenienced by the vehicle as discussed above. The model may be trained by using sensor data generated by a vehicle's perception system with labeled instances of when the vehicle is identified to be inconveniencing other traffic. The model may be trained to provide a value indicative of how likely the vehicle is inconveniencing the aforementioned road users. Alternatively, the model may be used to determine whether the vehicle is inconveniencing any other road users whenever the vehicle is stopped, regardless of the output of the various behavior models.

[0024] The output of the model may be used to determine an appropriate response. For instance, the value may be compared to a threshold value to determine whether the vehicle is inconveniencing other road users such that the vehicle should move. If the threshold is not met, the vehicle's computing devices may respond by keeping the vehicle at its current location. If the threshold is met, the vehicle's computing devices may respond by moving the vehicle in

order to reduce the value. In addition, how the vehicle is moved may depend on how the vehicle is inconveniencing other traffic. In yet another example, the vehicle's computing devices may attempt to search for a location with a lower value from the model.

[0025] The features described herein may enable autonomous vehicles to make independent determinations as to whether to move from a pick up location when waiting for a passenger to arrive. This may reduce or even eliminate the need for remote operators to make these decisions while also reducing the time it takes for the vehicle to react to such situations. This reduction in time may also increase the safety of the vehicle which may be able to respond much faster as the vehicle is able to continuously monitor its environment and anticipate inconvenience to other road users. At the same time, by allowing the vehicle to automatically determine that it is blocking or inconveniencing other road users, the vehicle may be more responsive to its environment and less likely to block, inconvenience, and in some cases, annoy, other road users.

Example Systems

[0026] As shown in FIG. **1**, a vehicle **100** in accordance with one aspect of the disclosure includes various components. While certain aspects of the disclosure are particularly useful in connection with specific types of vehicles, the vehicle may be any type of vehicle including, but not limited to, cars, trucks, motorcycles, busses, recreational vehicles, etc. The vehicle may have one or more computing devices, such as computing device **110** containing one or more processors **120**, memory **130** and other components typically present in general purpose computing devices.

[0027] The memory **130** stores information accessible by the one or more processors **120**, including instructions **132** and data **134** that may be executed or otherwise used by the processor **120**. The memory **130** may be of any type capable of storing information accessible by the processor, including a computing device-readable medium, or other medium that stores data that may be read with the aid of an electronic device, such as a hard-drive, memory card, ROM, RAM, DVD or other optical disks, as well as other write-capable and read-only memories. Systems and methods may include different combinations of the foregoing, whereby different portions of the instructions and data are stored on different types of media.

[0028] The instructions **132** may be any set of instructions to be executed directly (such as machine code) or indirectly (such as scripts) by the processor. For example, the instructions may be stored as computing device code on the computing device-readable medium. In that regard, the terms "instructions" and "programs" may be used interchangeably herein. The instructions may be stored in object code format for direct processing by the processor, or in any other computing device language including scripts or collections of independent source code modules that are interpreted on demand or compiled in advance. Functions, methods and routines of the instructions are explained in more detail below.

[0029] The data **134** may be retrieved, stored or modified by processor **120** in accordance with the instructions **132**. As an example, data **134** of memory **130** may store predefined scenarios. A given scenario may identify a set of scenario requirements including a type of object, a range of locations of the object relative to the vehicle, as well as other factors such as whether the autonomous vehicle is able to maneuver around the object, whether the object is using a turn signal, the condition of a traffic light relevant to the current location of the object, whether the object is approaching a stop sign, etc. The requirements may include discrete values, such as "right turn signal is on" or "in a right turn only lane", or ranges of values such as "having an heading that is oriented at an angle that is 30 to 60 degrees offset from a current path of vehicle **100**." In some examples, the predetermined scenarios may include similar information for multiple objects.

[0030] The one or more processor **120** may be any conventional processors, such as commercially available CPUs. Alternatively, the one or more processors may be a dedicated device such as an ASIC or other hardware-based processor. Although FIG. **1** functionally illustrates the processor, memory, and other elements of computing device **110** as being within the same block, it will be understood by those of ordinary skill in the art that the processor, computing device, or memory may actually include multiple processors, computing devices, or memories that may or may not be stored within the same physical housing. As an example, internal electronic display **152** may be controlled by a dedicated computing device having its own processor or central processing unit (CPU), memory, etc. which may interface with the computing device **110** via a high-bandwidth or other network connection. In some examples, this computing device may be a user interface computing device which can communicate with a user's client device. Similarly, the memory may be a hard drive or other storage media located in a housing different from that of computing device **110**. Accordingly, references to a processor or computing device will be understood to include references to a collection of processors or computing devices or memories that may or may not operate in parallel.

[0031] Computing device **110** may all of the components normally used in connection with a computing device such as the processor and memory described above as well as a user input **150** (e.g., a mouse, keyboard, touch screen and/or microphone) and various electronic displays (e.g., a monitor having a screen or any other electrical device that is operable to display information). In this example, the vehicle includes an internal electronic display **152** as well as one or more speakers **154** to provide information or audio visual experiences. In this regard, internal electronic display **152** may be located within a cabin of vehicle **100** and may be used by computing device **110** to provide information to passengers within the vehicle **100**. In addition to internal speakers, the one or more speakers **154** may include external speakers that are arranged at various locations on the vehicle in order to provide audible notifications to objects external to the vehicle **100**.

[0032] In one example, computing device **110** may be an autonomous driving computing system incorporated into vehicle **100**. The autonomous driving computing system may capable of communicating with various components of the vehicle. For example, returning to FIG. **1**, computing device **110** may be in communication with various systems of vehicle **100**, such as deceleration system **160** (for controlling braking of the vehicle), acceleration system **162** (for controlling acceleration of the vehicle), steering system **164** (for controlling the orientation of the wheels and direction of the vehicle), signaling system **166** (for controlling turn signals), navigation system **168** (for navigating the vehicle

to a location or around objects), positioning system **170** (for determining the position of the vehicle), perception system **172** (for detecting objects in the vehicle's environment), and power system **174** (for example, a battery and/or gas or diesel powered engine) in order to control the movement, speed, etc. of vehicle **100** in accordance with the instructions **132** of memory **130** in an autonomous driving mode which does not require or need continuous or periodic input from a passenger of the vehicle. Again, although these systems are shown as external to computing device **110**, in actuality, these systems may also be incorporated into computing device **110**, again as an autonomous driving computing system for controlling vehicle **100**.

[0033] The computing device **110** may control the direction and speed of the vehicle by controlling various components. By way of example, computing device **110** may navigate the vehicle to a destination location completely autonomously using data from the map information and navigation system **168**. Computing devices **110** may use the positioning system **170** to determine the vehicle's location and perception system **172** to detect and respond to objects when needed to reach the location safely. In order to do so, computing devices **110** may cause the vehicle to accelerate (e.g., by increasing fuel or other energy provided to the engine by acceleration system **162**), decelerate (e.g., by decreasing the fuel supplied to the engine, changing gears, and/or by applying brakes by deceleration system **160**), change direction (e.g., by turning the front or rear wheels of vehicle **100** by steering system **164**), and signal such changes (e.g., by lighting turn signals of signaling system **166**). Thus, the acceleration system **162** and deceleration system **160** may be a part of a drivetrain that includes various components between an engine of the vehicle and the wheels of the vehicle. Again, by controlling these systems, computing devices **110** may also control the drivetrain of the vehicle in order to maneuver the vehicle autonomously.

[0034] As an example, computing device **110** may interact with deceleration system **160** and acceleration system **162** in order to control the speed of the vehicle. Similarly, steering system **164** may be used by computing device **110** in order to control the direction of vehicle **100**. For example, if vehicle **100** configured for use on a road, such as a car or truck, the steering system may include components to control the angle of wheels to turn the vehicle. Signaling system **166** may be used by computing device **110** in order to signal the vehicle's intent to other drivers or vehicles, for example, by lighting turn signals or brake lights when needed.

[0035] Navigation system **168** may be used by computing device **110** in order to determine and follow a route to a location. In this regard, the navigation system **168** and/or data **134** may store map information, e.g., highly detailed maps that computing devices **110** can use to navigate or control the vehicle. As an example, these maps may identify the shape and elevation of roadways, lane markers, intersections, crosswalks, speed limits, traffic signal lights, buildings, signs, real time or historical traffic information, vegetation, or other such objects and information. The lane markers may include features such as solid or broken double or single lane lines, solid or broken lane lines, reflectors, etc. A given lane may be associated with left and right lane lines or other lane markers that define the boundary of the lane. Thus, most lanes may be bounded by a left edge of one lane line and a right edge of another lane line. As noted above, the map information may store known traffic or congestion information and/or and transit schedules (train, bus, etc.) from a particular pickup location at similar times in the past. This information may even be updated in real time by information received by the computing devices **110**.

[0036] FIG. **2** is an example of map information **200** for a section of roadway including intersection **220**. In this example, map information **200** depicts a portion of the map information that includes information identifying the shape, location, and other characteristics of lane markers or lane lines **210**, **212**, **214**, stop signs **240**, **242**, **244**, **246**, as well as stop lines **250**, **252**, **254**, **256**. The lane lines may also define various lanes **260**, **261**, **264**, **265**, implied by the width of a road, such as lanes **262**, **263**, **266**, **267**, and/or lanes **260-267** may also be explicitly identified in the map information **200**. In this example, map information **200** also identifies bicycle lanes **280**, **282**. In addition to these features, the map information may also include information that identifies the direction of traffic and speed limits for each lane as well as information that allows the computing devices **110** to determine whether the vehicle has the right of way to complete a particular maneuver (i.e. complete a turn or cross a lane of traffic or intersection), as well as other features such as curbs, buildings, waterways, vegetation, signs, etc.

[0037] The map information may also store predetermined stopping locations including pickup locations and drop off locations. Pickup locations may refer to locations where the autonomous vehicle stops to wait to pick up a passenger for a trip. Drop off locations may refer to locations where the autonomous vehicle stops to allow for a passenger to exit the vehicle after a trip. Other stopping locations, such as those where the vehicle stops, allows the passenger to exit and waits for the passenger to return are also possible. Each of these may simply be parking locations that are discrete, pre-determined pickup and drop off locations, and in some instances, hand-selected by a human operator or learned by a computing device over time. In this regard, each stopping location may be a locations where a vehicle could stop selected through some manual or automated analysis of the characteristics of each location. In the example of FIG. **2**, the map information includes stopping locations **290**, **292**. However, in some locations, such as in neighborhoods or narrow roads, as in the example of FIG. **2**, there may be few or no "fixed" or predesignated parking spaces or areas.

[0038] Although the detailed map information is depicted herein as an image-based map, the map information need not be entirely image based (for example, raster). For example, the detailed map information may include one or more roadgraphs or graph networks of information such as roads, lanes, intersections, and the connections between these features. Each feature may be stored as graph data and may be associated with information such as a geographic location and whether or not it is linked to other related features, for example, a stop sign may be linked to a road and an intersection, etc. In some examples, the associated data may include grid-based indices of a roadgraph to allow for efficient lookup of certain roadgraph features.

[0039] The perception system **172** also includes one or more components for detecting objects external to the vehicle such as other vehicles, obstacles in the roadway, traffic signals, signs, trees, etc. For example, the perception system **172** may include one or more LIDAR sensors, sonar devices, radar units, cameras and/or any other detection

devices that record data which may be processed by computing devices **110**. The sensors of the perception system may detect objects and their characteristics such as location, orientation, size, shape, type (for instance, vehicle, pedestrian, bicyclist, etc.), heading, and speed of movement, etc. The raw data from the sensors and/or the aforementioned characteristics can be quantified or arranged into a descriptive function, vector, and or bounding box and sent for further processing to the computing devices **110** periodically and continuously as it is generated by the perception system **172**. As discussed in further detail below, computing devices **110** may use the positioning system **170** to determine the vehicle's location and perception system **172** to detect and respond to objects when needed to reach the location safely.

[0040] For instance, FIG. **3** is an example external view of vehicle **100**. In this example, roof-top housing **310** and dome housing **312** may include a LIDAR sensor as well as various cameras and radar units. In addition, housing **320** located at the front end of vehicle **100** and housings **330**, **332** on the driver's and passenger's sides of the vehicle may each store a LIDAR sensor. For example, housing **330** is located in front of driver door **350**. Vehicle **100** also includes housings **340**, **342** for radar units and/or cameras also located on the roof of vehicle **100**. Additional radar units and cameras (not shown) may be located at the front and rear ends of vehicle **100** and/or on other positions along the roof or roof-top housing **310**. Vehicle **100** also includes many features of a typical passenger vehicle such as doors **350**, **352**, wheels **360**, **362**, etc.

[0041] Once a nearby object is detected, computing devices **110** and/or perception system **172** may determine the object's type, for example, a traffic cone, pedestrian, a vehicle (such as a passenger car, truck, bus, etc.), bicycle, etc. Objects may be identified by various models which may consider various characteristics of the detected objects, such as the size of an object, the speed of the object (bicycles do not tend to go faster than 40 miles per hour or slower than 0.1 miles per hour), the heat coming from the bicycle (bicycles tend to have rider that emit heat from their bodies), etc. In addition, the object may be classified based on specific attributes of the object, such as information contained on a license plate, bumper sticker, or logos that appear on the vehicle.

[0042] In one example, computing devices **110** may be operable to predict another vehicle's future movement based solely on the other vehicle's instant direction, acceleration/deceleration and velocity, e.g., that the other vehicle's current direction and movement will continue. However, memory **130** may also store behavior models that provide the probability of one or more actions being taken a detected object. To increase the usefulness of these behavior models, each behavior model may be associated with a specific type of object. For instance, one type of behavior model may be used for objects identified as pedestrians, another type of behavior model may be used for objects identified as vehicles, another type of behavior may be used for objects identified as bicycles or bicyclists, etc. The behavior models may be used by the computing devices **110** in order to predict a detected object future movement by analyzing data relating to the other vehicle's current surroundings (such as the detected or estimated size, shape, location, orientation, heading, velocity, acceleration or deceleration, change in acceleration or deceleration, etc.), and determining how that other object will likely respond to those surroundings. In this regard, the behavior models may function from an object-centric, view of the object's environment, in that the system determines what the other objects are perceiving in order to better predict how those objects will behave. In this regard, in at least some instances, the behavior models may also indicate whether the predicted behavior for an object is responsive to a particular other object including the vehicle **100**.

[0043] The one or more computing devices **110** of vehicle **100** may also receive or transfer information to and from other computing devices, for instance using wireless network connections **156**. The wireless network connections may include, for instance, BLUETOOTH®, Bluetooth LE, LTE, cellular, near field communications, etc. and various combinations of the foregoing. FIGS. **4** and **5** are pictorial and functional diagrams, respectively, of an example system **400** that includes a plurality of computing devices **410**, **420**, **430**, **440** and a storage system **450** connected via a network **460**. System **400** also includes vehicle **100**, and vehicle **100**A which may be configured similarly to vehicle **100**. Although only a few vehicles and computing devices are depicted for simplicity, a typical system may include significantly more.

[0044] As shown in FIG. **4**, each of computing devices **410**, **420**, **430**, **440** may include one or more processors, memory, data and instructions. Such processors, memories, data and instructions may be configured similarly to one or more processors **120**, memory **130**, data **134**, and instructions **132** of computing device **110**.

[0045] The network **460**, and intervening nodes, may include various configurations and protocols including short range communication protocols such as BLUETOOTH®, Bluetooth LE, the Internet, World Wide Web, intranets, virtual private networks, wide area networks, local networks, private networks using communication protocols proprietary to one or more companies, Ethernet, WiFi and HTTP, and various combinations of the foregoing. Such communication may be facilitated by any device capable of transmitting data to and from other computing devices, such as modems and wireless interfaces.

[0046] In one example, one or more computing devices **110** may include a server having a plurality of computing devices, e.g., a load balanced server farm, that exchange information with different nodes of a network for the purpose of receiving, processing and transmitting the data to and from other computing devices. For instance, one or more computing devices **410** may include one or more server computing devices that are capable of communicating with one or more computing devices **110** of vehicle **100** or a similar computing device of vehicle **100**A as well as client computing devices **420**, **430**, **440** via the network **460**. For example, vehicles **100** and **100**A may be a part of a fleet of vehicles that can be dispatched by server computing devices to various locations. In this regard, the vehicles of the fleet may periodically send the server computing devices location information provided by the vehicle's respective positioning systems and the one or more server computing devices may track the locations of the vehicles.

[0047] In addition, server computing devices **410** may use network **460** to transmit and present information to a user, such as user **422**, **432**, **442** on a display, such as displays **424**, **434**, **444** of computing devices **420**, **430**, **440**. In this regard, computing devices **420**, **430**, **440** may be considered client computing devices.

[0048] As shown in FIG. **5**, each client computing device **420**, **430**, **440** may be a personal computing device intended for use by a user **422**, **432**, **442**, and have all of the components normally used in connection with a personal computing device including a one or more processors (e.g., a central processing unit (CPU)), memory (e.g., RAM and internal hard drives) storing data and instructions, a display such as displays **424**, **434**, **444** (e.g., a monitor having a screen, a touch-screen, a projector, a television, or other device that is operable to display information), and user input devices **426**, **436**, **446** (e.g., a mouse, keyboard, touchscreen or microphone). The client computing devices may also include a camera for recording video streams, speakers, a network interface device, and all of the components used for connecting these elements to one another.

[0049] Although the client computing devices **420**, **430**, and **440** may each comprise a full-sized personal computing device, they may alternatively comprise mobile computing devices capable of wirelessly exchanging data with a server over a network such as the Internet. By way of example only, client computing device **420** may be a mobile phone or a device such as a wireless-enabled PDA, a tablet PC, a wearable computing device or system, or a netbook that is capable of obtaining information via the Internet or other networks. In another example, client computing device **430** may be a wearable computing system, shown as a wrist watch in FIG. **4**. As an example the user may input information using a small keyboard, a keypad, microphone, using visual signals with a camera, or a touch screen.

[0050] In some examples, client computing device **440** may be remote operator work station used by an administrator to provide remote operator services to users such as users **422** and **432**. For example, a remote operator **442** may use the remote operator work station **440** to communicate via a telephone call or audio connection with users through their respective client computing devices and/or vehicles **100** or **100**A in order to ensure the safe operation of vehicles **100** and **100**A and the safety of the users as described in further detail below. Although only a single remote operator work station **440** is shown in FIGS. **4** and **5**, any number of such work stations may be included in a typical system.

[0051] Storage system **450** may store various types of information as described in more detail below. This information may be retrieved or otherwise accessed by a server computing device, such as one or more server computing devices **410**, in order to perform some or all of the features described herein. For example, the information may include user account information such as credentials (e.g., a username and password as in the case of a traditional single-factor authentication as well as other types of credentials typically used in multi-factor authentications such as random identifiers, biometrics, etc.) that can be used to identify a user to the one or more server computing devices. The user account information may also include personal information such as the user's name, contact information, identifying information of the user's client computing device (or devices if multiple devices are used with the same user account), as well as age information, health information, and user history information about how long it has taken the user to enter or exit vehicles in the past as discussed below.

[0052] The storage system **450** may also store routing data for generating and evaluating routes between locations. For example, the routing information may be used to estimate how long it would take a vehicle at a first location to reach a second location. In this regard, the routing information may include map information, not necessarily as particular as the detailed map information described above, but including roads, as well as information about those road such as direction (one way, two way, etc.), orientation (North, South, etc.), speed limits, as well as traffic information identifying expected traffic conditions, etc.

[0053] The storage system **450** may also store information which can be provided to client computing devices for display to a user. For instance, the storage system **450** may store predetermined distance information for determining an area at which a vehicle is likely to stop for a given pickup or destination location. The storage system **450** may also store graphics, icons, and other items which may be displayed to a user as discussed below.

[0054] As with memory **130**, storage system **450** can be of any type of computerized storage capable of storing information accessible by the server computing devices **410**, such as a hard-drive, memory card, ROM, RAM, DVD, CD-ROM, write-capable, and read-only memories. In addition, storage system **450** may include a distributed storage system where data is stored on a plurality of different storage devices which may be physically located at the same or different geographic locations. Storage system **450** may be connected to the computing devices via the network **460** as shown in FIG. **4** and/or may be directly connected to or incorporated into any of the computing devices **110**, **410**, **420**, **430**, **440**, etc.

Example Methods

[0055] In addition to the operations described above and illustrated in the figures, various operations will now be described. It should be understood that the following operations do not have to be performed in the precise order described below. Rather, various steps can be handled in a different order or simultaneously, and steps may also be added or omitted.

[0056] In one aspect, a user may download an application for requesting a vehicle to a client computing device. For example, users **422** and **432** may download the application via a link in an email, directly from a website, or an application store to client computing devices **420** and **430**. For example, client computing device may transmit a request for the application over the network, for example, to one or more server computing devices **410**, and in response, receive the application. The application may be installed locally at the client computing device.

[0057] The user may then use his or her client computing device to access the application and request a vehicle. As an example, a user such as user **432** may use client computing device **430** to send a request to one or more server computing devices **410** for a vehicle. As part of this, the user may identify a pickup location, a destination location, and, in some cases, one or more intermediate stopping locations anywhere within a service area where a vehicle can stop.

[0058] These pickup and destination locations may be predefined (e.g., specific areas of a parking lot, etc.) or may simply be any location within a service area of the vehicles. As an example, a pickup location can be defaulted to the current location of the user's client computing device, or can be input by the user at the user's client device. For instance, the user may enter an address or other location information or select a location on a map to select a pickup location. Once the user has selected one or more of a pickup and/or

destination locations, the client computing device **420** may send the location or locations to one or more server computing devices of the centralized dispatching system. In response, one or more server computing devices, such as server computing device **410**, may select a vehicle, such as vehicle **100**, for instance based on availability and proximity to the user. The server computing device **410** may then assign the user as the passenger for the vehicle **100**, dispatch the selected vehicle (here vehicle **100**) to pick up to the assigned passenger. This may include by providing the vehicle's computing devices **110** with the pickup and/or destination locations specified by the assigned passenger as well as information that can be used by the computing devices **110** of vehicle **100** to authenticate the client computing device, such as client computing device **430**.

[0059] Thereafter, the computing devices **110** may maneuver vehicle **100** towards the pickup location and, after picking up the passenger, towards the destination location. Once the vehicle is within a predetermined distance of a pickup or drop off location, such as 50 meters or more or less, the computing devices **110** may begin to look for a place to stop and allow a passenger to enter or exit the vehicle. In some instances, the vehicle may pull towards a curb partially in a lane, in a lane, in front of a driveway or parking spot. While it may be appropriate in some circumstances to stop in such locations, this may cause inconvenience to other road users.

[0060] FIG. **6** is an example **600** bird's eye view of a section of roadway corresponding to map information **200**. In this example, the shape location and other characteristics of intersection **620**, lane lines **610**, **612**, **614**, stop signs **640**, **642**, **644**, **646**, stop lines **650**, **652**, **654**, **656**, lanes **660-268**, bicycle lanes **680**, **682**, and parking spots **690**, **692** may generally correspond to the shape location and other characteristics of intersection **220**, lane lines **210**, **212**, **214**, stop signs **240**, **242**, **244**, **246**, stop lines **250**, **252**, **254**, **256**, lanes **260-268**, bicycle lanes **280**, **282**, and stopping locations **290**, **292**, respectively. FIGS. **7-10**, discussed further below provide examples situations in which a vehicle can block or inconvenience other vehicles when stopped and waiting for a passenger using the example **600** of FIG. **6**. Although these examples are useful for demonstration purposes, they should not be considered limiting; a vehicle may block or inconvenience other road users in an infinite number of ways.

[0061] For instance, turning to FIG. **7**, vehicle **100** may be stopped in lane **662** adjacent to a pickup location **710** and waiting for a passenger. In this example, a second vehicle **720** may be approaching vehicle **100** from behind. In this regard, vehicle **100** may be blocking and inconveniencing second vehicle **720**. As shown in FIG. **8**, vehicle **100** may be stopped in lane **663** adjacent to a pickup location **810** and waiting for a passenger. In this example, a second vehicle **820** may be approaching vehicle **100** from an adjacent lane **662**. In this regard, vehicle **100** may be blocking and inconveniencing second vehicle **820**. Turning to FIG. **9**, vehicle **100** may be stopped in lane **661** adjacent to a pickup location **910** and waiting for a passenger. In this example, a second vehicle **920** may be parked in parking spot **690** and attempting to "unpark" or rather, waiting to pull out into lane **661**. In this regard, vehicle **100** may be blocking and inconveniencing second vehicle **920**. FIG. **10** provides a similar example of vehicle **100** stopped adjacent to a pickup location **1030** and blocking a second vehicle **1040** from turning into and reaching an adjacent area **1050** corresponding to a driveway. In this regard, vehicle **100** may be blocking and inconveniencing second vehicle **1040**. As shown in FIG. **11**, vehicle **100** may be stopped in lane **660** adjacent to a pickup location **1110** and waiting for a passenger. In this example, vehicle **100** is at least partially blocking the bicycle lane **680**, and a bicyclist **1120** is approaching vehicle **100** from behind.

[0062] As one approach, when stopped, computing devices **110** may request assistance at consistent, increasing and/or decreasing time intervals from remote operators as to whether the vehicle should move along based on whether the vehicle is blocking or otherwise inconveniencing other road users. For instance, the computing devices **110** may send a request for assistance via network **460** to remote operator **442**'s remote operator work station **440**. This request may include one or more still and/or video images of the vehicle's surroundings captured by the perception system **172**. In response, the remote operator **442** may review the images of the vehicle's surroundings and instruct the vehicle's computing devices **110** to stay put or move along depending on the flow of traffic around the vehicle **100**. Of course, this back-and-forth over the network **460** can take time, which can cause annoyance and inconvenience to other road users. To reduce or even eliminate the need for the autonomous vehicles to call a remote operator in such situations, the computing devices **110** may automatically identify situations in which the vehicle is inconveniencing other drivers and thereafter attempt to reduce that inconvenience.

[0063] In order to do so, when stopped, the computing devices **110** may receive sensor data from the vehicle's perception system **172**. This sensor data may then be used to determine whether the vehicle is inconveniencing other road users, such as by blocking traffic or causing other road users to have to move around the vehicle. For instance, the computing devices **110** may determine whether the vehicle is preventing other road users from proceeding or causing those road users to have to drive around the vehicle as discussed further below. As an example, the computing devices may determine whether another vehicle has stopped close behind the vehicle, in other words, so as to be "tailgating."

[0064] In addition or alternatively, as noted above, the computing devices **110** may receive output from various behavior prediction models. For instance, for each road user of a given type identified from the sensor data provided by the perception system **172**, the vehicle's computing devices may use a corresponding behavior prediction model for that type (for instance, vehicle, pedestrian, bicyclist, etc.) to output a future behavior or trajectory for that road user. In some instances, these behavior models may also identify whether the other road user's behavior or trajectory is responsive to the vehicle being slowed or stopped in its present location.

[0065] If the predictions change over time or if a prediction does not sufficiently correspond with the actual behavior of the other road user as the other road user approaches the vehicle and these changes or differences are caused by (i.e. in response to) the vehicle, this may indicate that the vehicle is blocking or inconveniencing the other road user. As an example, if the predicted behavior (or trajectory) includes the other road user proceeding in its lane adjacent to the vehicle or directly through the location of the vehicle, but the actual, observed behavior (or trajectory) has the other road user moving away from the vehicle (laterally) or

stopping behind the vehicle, this may indicate that the vehicle is blocking or inconveniencing the other road user.

[0066] For instance, FIG. **12** depicts a predicted trajectory **1210** and an actual trajectory **1220** of a road user with respect to vehicle **100** when stopped, for instance as shown in the example of FIG. **7**. In the example of FIG. **12**, the predicted trajectory **1210** passes through the location of vehicle **100**, but the actual trajectory **1220** has the other road user moving laterally away from the vehicle. FIG. **13** depicts a predicted trajectory **1210** and an actual trajectory **1320** of a road user with respect to vehicle **100** when stopped, for instance as shown in the example of FIG. **7**. In the example of FIG. **13**, the predicted trajectory **1310** passes through the location of vehicle **100**, but the actual trajectory **1320** has the other road user stopping behind vehicles **100**. The magnitude of this difference as well as how long another road user must for the vehicle to move may also indicate how much the vehicle is inconveniencing the other road user or whether the vehicle is completely blocking the other road user.

[0067] Once the vehicle is determined to be inconveniencing another road user while stopped, the computing devices **110** may evaluate or determine a value for that inconvenience. In order to allow the computing devices **110** to quickly assess the inconvenience to another road user, a model may be used to provide an inconvenience value for only those other road users detected by the vehicle's perception system as being blocked or inconvenienced by the vehicle as discussed above. For instance, the model may include a machine learning model, such as a deep neural network or other types of machine learning models.

[0068] The model may be trained "offline", for instance, by one or more of server computing devices **410**, and sent via network **460** or otherwise downloaded to memory **130** of computing devices **110** for later use. The training data for the model may include using sensor data generated by a vehicle's perception system with labeled instances including locations the vehicle is identified to be blocking and/or inconveniencing other traffic. These labels may be generated, for instance, based on the aforementioned determinations by remote operators for different vehicles and further labeling those determinations with an inconvenience value. This inconvenience value may be a value selected by the remote operator on some predefined scale, such as 0 to 1 or 1 to 10, etc. when the remote operator is reviewing the images of a vehicle's environment. The sensor data may include all of the data generated by the corresponding vehicle's perception system, or more streamlined information, such as the pose, velocity, acceleration and similar data for each road user detected by the perception system over some period of time. The model may be trained using the training data in order to provide a value indicative of how likely the vehicle is blocking and/or inconveniencing the aforementioned road users when the vehicle is stopped at a given location (typically the location where the vehicle is currently stopped) and thereafter sent to the vehicles of the fleet.

[0069] Alternatively, the model may be used to determine whether the vehicle is inconveniencing any other road users whenever the vehicle is stopped, regardless of the output of the various behavior models.

[0070] The output of the model may be used to determine an appropriate response. For instance, the value may be compared to a threshold value to determine whether the vehicle blocking and/or inconveniencing other road users such that the vehicle should move. As an example, if the value is on a scale of 0 to 1, where 1 is very high inconvenience to the other road user, there may be a predetermined threshold somewhere in between (for example, 0.7 or more or less). When the threshold is not met, the vehicle's computing devices, the computing devices may determine that vehicle **100** is not inconveniencing another road user. In this case, the computing devices **110** may respond by keeping the vehicle **100** at its current location. When the threshold is met, the vehicle's computing devices determine that the vehicle is inconveniencing another road user. In this case, the computing devices **110** may respond by moving the vehicle in order to reduce the value.

[0071] How the vehicle is moved may depend on how the vehicle is blocking or inconveniencing other traffic. For instance, based on the relative position of the road user identified as being inconvenienced by the model, the computing devices may choose a different type of movement. For example, if the vehicle is inconveniencing a road user in an adjacent lane such as in the example of FIG. **8**, the computing devices **110** may move the vehicle forward and/or to the right (in a right-hand drive country). As another example, if the vehicle is inconveniencing another road user that is behind the vehicle such as in the example of FIG. **7**, the vehicle's computing devices may attempt to move the vehicle further to the right (in a right-hand drive country) in order to allow the another road user to pass the vehicle on the left. These movements, in turn, may actually reduce the value.

[0072] In yet another example, the vehicle's computing devices may attempt to search for a new location with a lower value using the model. For instance, the vehicle's computing devices may search nearby stopping locations that are further out the way of the road user and not blocking where the vehicle's computing devices predict that the road user is attempting to go, potentially at the expense of being less convenient to the passenger that is being picked up. The computing devices **110** may then input these new locations and sensor data from the perception system into the model in order to determine an inconvenience value for the new locations. The new location with the lowest value may then be selected. Alternatively, the map may include data identifying an estimated inconvenience value for different stopping locations. In this regard, the computing devices may simply search and select new location with a lower inconvenience value than the vehicle's current location.

[0073] Once a new location is selected, the computing devices **110** may control the vehicle in order to move to that lower value location. If the computing devices **110** are unable to find a location with a lower value, the vehicle's computing devices may send a message to a passenger of the vehicle (for example, to the passenger's client computing device for display) indicating that the vehicle is going to move to a new location. The vehicle's computing devices may also control the vehicle in order to proceed forward. At the same time, the vehicle's computing devices may attempt to identify this new location, for instance, by searching over pre-mapped pull over locations or attempting to identify nearby parking areas that are within some distance of the vehicle's current location.

[0074] FIG. **14** is a flow diagram **1400** that may be performed by one or more processors such as one or more processors **120** of computing device **110**. In this example, at block **1410**, a vehicle having an autonomous driving mode

is stopped at a first location. At block **1420**, while the vehicle is stopped, sensor data is received from a perception system of the vehicle. This sensor data identifiers a road user. At block **1430**, using the sensor data, a value indicative of a level of inconvenience to the road user caused by stopping the vehicle at the first location is determined. At block **1440**, the vehicle is controlled in the autonomous driving mode in order to move the vehicle from the first location in order to reduce the value.

[0075] The features described herein may enable autonomous vehicles to make independent determinations as to whether to move from a pick up location when waiting for a passenger to arrive. This may reduce or even eliminate the need for remote operators to make these decisions while also reducing the time it takes for the vehicle to react to such situations. This reduction in time may also increase the safety of the vehicle which may be able to respond much faster as the vehicle is able to continuously monitor its environment and anticipate inconvenience to other road users. At the same time, by allowing the vehicle to automatically determine that it is blocking or inconveniencing other road users, the vehicle may be more responsive to its environment and less likely to block, inconvenience, and in some cases, annoy, other road users.

[0076] Unless otherwise stated, the foregoing alternative examples are not mutually exclusive, but may be implemented in various combinations to achieve unique advantages. As these and other variations and combinations of the features discussed above can be utilized without departing from the subject matter defined by the claims, the foregoing description of the embodiments should be taken by way of illustration rather than by way of limitation of the subject matter defined by the claims. In addition, the provision of the examples described herein, as well as clauses phrased as "such as," "including" and the like, should not be interpreted as limiting the subject matter of the claims to the specific examples; rather, the examples are intended to illustrate only one of many possible embodiments. Further, the same reference numbers in different drawings can identify the same or similar elements.

1. A method of reducing inconvenience to a road user caused by an autonomous vehicle, the method comprising:

identifying, by one or more processors of the autonomous vehicle, a relative position of the road user to a location at which the autonomous vehicle is currently stopped;

determining, by the one or more processors of the autonomous vehicle, based on the relative position of the road user with respect to the location at which the autonomous vehicle is currently stopped, that the autonomous vehicle is inconveniencing the road user; and

controlling, by the one or more processors, the autonomous vehicle to perform a maneuver based on the determination that the autonomous vehicle is inconveniencing the road user and the relative position of the road user with respect to the location at which the autonomous vehicle is currently stopped.

2. The method of claim **1**, wherein the relative position of the road user is in a lane adjacent to the location at which the autonomous vehicle is currently stopped.

3. The method of claim **1**, wherein the relative position of the road user is behind the location at which the autonomous vehicle is currently stopped.

4. The method of claim **1**, further comprising generating, by the one or more processors, an inconvenience value using a behavior prediction model and sensor data identifying the road user, the inconvenience value indicating how much the autonomous vehicle is inconveniencing the road user, and wherein determining that the location at which the autonomous vehicle is currently stopped is inconveniencing the road user includes comparing the inconvenience value to a threshold value.

5. The method of claim **4**, wherein the autonomous vehicle is inconveniencing the road user when the inconvenience value is equal to or greater than the threshold value.

6. The method of claim **4**, wherein the inconvenience value is on a scale of 0 to 1.

7. The method of claim **4**, wherein when the inconvenience value is less than the threshold value, the autonomous vehicle waits at the location at which the autonomous vehicle is currently stopped for a passenger to arrive.

8. The method of claim **4**, wherein controlling the autonomous vehicle is performed in order to reduce the inconvenience value.

9. An autonomous vehicle comprising one or more processors configured to:

identify a relative position of a road user to a location at which the autonomous vehicle is currently stopped;

determine, based on the relative position of the road user with respect to the location at which the autonomous vehicle is currently stopped, that the autonomous vehicle is inconveniencing the road user; and

control the autonomous vehicle to perform a maneuver based on the determination that the autonomous vehicle is inconveniencing the road user and the relative position of the road user with respect to the location at which the autonomous vehicle is currently stopped.

10. The autonomous vehicle of claim **9**, wherein the relative position of the road user is in a lane adjacent to the location at which the autonomous vehicle is currently stopped.

11. The autonomous vehicle of claim **9**, wherein the relative position of the road user is behind the location at which the autonomous vehicle is currently stopped.

12. The autonomous vehicle of claim **9**, wherein the one or more processors are further configured to generate an inconvenience value using a behavior prediction model and sensor data identifying the road user, the inconvenience value indicating how much the autonomous vehicle is inconveniencing the road user, and compare the inconvenience value to a threshold value.

13. The autonomous vehicle of claim **12**, wherein the autonomous vehicle is inconveniencing the road user when the inconvenience value is equal to or greater than the threshold value.

14. The autonomous vehicle of claim **12**, wherein the inconvenience value is on a scale of 0 to 1.

15. The autonomous vehicle of claim **12**, wherein when the inconvenience value is less than the threshold value, the autonomous vehicle waits at the location at which the autonomous vehicle is currently stopped for a passenger to arrive.

16. The autonomous vehicle of claim **12**, wherein the autonomous vehicle is controlled in order to reduce the inconvenience value.

17. A non-transitory, computer-readable medium storing instructions which, when executed, cause a computing

device to perform a method of reducing inconvenience to a road user caused by an autonomous vehicle, the method comprising:

identifying a relative position of the road user to a location at which the autonomous vehicle is currently stopped;

determining, based on the relative position of the road user with respect to the location at which the autonomous vehicle is currently stopped, that the autonomous vehicle is inconveniencing the road user; and

controlling the autonomous vehicle to perform a maneuver based on the determination that the autonomous vehicle is inconveniencing the road user and the relative position of the road user with respect to the location at which the autonomous vehicle is currently stopped.

18. The medium of claim **17**, wherein the relative position of the road user is in a lane adjacent to the location at which the autonomous vehicle is currently stopped.

19. The medium of claim **17**, wherein the relative position of the road user is behind the location at which the autonomous vehicle is currently stopped.

20. The medium of claim **17**, the method further comprising generating an inconvenience value using a behavior prediction model and sensor data identifying the road user, the inconvenience value indicating how much the autonomous vehicle is inconveniencing the road user, and wherein determining that the location at which the autonomous vehicle is currently stopped is inconveniencing the road user includes comparing the inconvenience value to a threshold value.

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